SWARIT TIWARI

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SUMMARY

M.Sc. Mechanical Engineering & Management graduate with experience in CAD/CAE design, manufacturing improvement, data analysis and business-oriented engineering projects. Skilled in mechanical design, simulation, Python automation and analytical tools, with project work across product development, maritime fuels and NLP-based customer insights.

SKILLS

Languages: German (B2), English (Fluent), Hindi (Native)

Engineering Software: CATIA 3DEXPERIENCE, Siemens NX, SolidWorks, Fusion 360, AutoCAD, ANSYS, Inventor

Business Tools: Power BI, Tableau, Jira, SAP ERP, Advanced Excel, MS Office

Programming: Python (Pandas, NumPy, transformers), MATLAB, Simulink, SQL.

EDUCATION

10/2022 – 03/2026 **M.Sc. in Mechanical Engineering and Management**
Technische Universität Hamburg, Germany

07/2017 – 07/2021 **B.Eng. in Mechanical Engineering**
Gujarat Technological University, India

PROFESSIONAL EXPERIENCE

07/2021 – 08/2022 **Jr. Design Engineer**
GERENT, India
- Developed and validated mechanical components using Fusion 360, SolidWorks and ANSYS, supporting design, simulation and production readiness.
- Analyzed manufacturing processes and implemented improvements that reduced production line waste by 35%.
- Automated technical certification documentation with Python/VBA, reducing documentation effort by 75%.

06/2020 – 07/2020 **Intern**
GURUKRUPA KRAFTS PVT. LTD., India
- Operated and maintained production machinery, including lathe, CNC, VMC, gear cutters, and hydraulic presses, gaining in-depth manufacturing knowledge.

PROJECTS

Sentiment Analysis and Topic Modelling of Restaurant Reviews | Master Thesis | Python, BERTopic, BERT
Filtered 1M+ Google reviews to 212k+ restaurant reviews and extracted customer-experience insights using NLP.

Alternative Fuels in Maritime Industry | Fraunhofer CML Project | Power BI, Jira, Excel
Developed a weighted scoring matrix to compare alternative maritime fuels across commercial and operational criteria.

Single-Print FDM Race Car | Fusion 360, SolidWorks, FDM, DfAM
Designed a movable single-print race car with functional wheels, printability and no-assembly design.

Topology-Optimized 3D Printed Bridge | SolidWorks, Simulation, FFF/MEX, Testing
Developed and tested a lightweight bridge, 172.6 g mass and 500+ N load capacity.

Contactless Sanitization Station | CAD Concept, Product Design, Autodesk Hackathon
Developed a CAD concept for an automatic sanitization station, awarded 2nd place at Autodesk Hackathon.

Regenerative Braking System for Bicycles | CAD, ANSYS, Mechanical Design
Designed and validated a modular front-wheel regenerative braking concept for bicycles.